

Wideband Planer Folded Dipole Antenna for Fast Wireless Communication.

Mayur Borse

UG Students, Department of EXTC,
K.J. Somaiya Institute of Technology,
University of Mumbai, India
m.borse@somaiya.edu

Jayesh Dhandge

UG Students, Department of EXTC,
K.J. Somaiya Institute of Technology,
University of Mumbai, India
jayesh.dhandge@somaiya.edu

Yash Dhekale

UG Students, Department of EXTC,
K.J. Somaiya Institute of Technology,
University of Mumbai, India
yash.dhekale@somaiya.edu

Prof. Martand Jha

Asst. Professor, Department of EXTC,
K.J. Somaiya Institute of Technology,
University of Mumbai, India
mjha@somaiya.edu

Dr. Kiran Rathod

Asst. Professor, Department of EXTC,
K.J. Somaiya Institute of Technology,
University of Mumbai, India
kiran.rathod@somaiya.edu

Abstract—This paper presents the design and analysis of a Wideband Planer Folded Dipole Antenna for Fast Wireless Communication operating in the 5-6 GHz frequency range using HFSS. The Proposed antenna is designed on FR4 substrate having thickness 0.6mm. The proposed antenna exhibits impedance bandwidth of 500 MHz (5.38 - 5.8 MHz, 7.64%) and it has a gain of 5 dB and a narrow beamwidth ($30^\circ \times 30^\circ$), making it suitable for modern wireless communication and radar applications. It has return loss of -22 dB at 5.5 GHz resonance frequency. Proposed antenna is ensuring optimal performance in terms of radiation characteristics, impedance matching, and power handling capacity. Measured and simulated results validate the antenna's efficiency, showcasing a stable radiation pattern, low return loss, and desirable gain at desired frequency band. The proposed design demonstrates significant improvements over conventional broadband antennas in terms of compactness and directivity, making it a promising candidate for high-frequency applications.

Keywords—Folded dipole antenna, ultra-wideband (UWB), high gain, wireless communication and radar applications, antenna design, electromagnetic simulation, return loss.

I. INTRODUCTION

The increasing demand for high-speed wireless communication and precision radar systems has driven the need for antennas that combine high gain, broad bandwidth, and compact dimensions. These attributes are particularly vital in emerging applications such as 5G communication, satellite links, unmanned aerial vehicles (UAVs), and portable military-grade RF systems. As antennas serve as the core interface for transmitting and receiving electromagnetic signals, their design has a direct impact on system performance in terms of range, signal quality, and data throughput.

Over the years, several broadband antenna structures have been investigated to meet these evolving requirements. Designs such as the Log-Periodic Dipole Antenna (LPDA), Vivaldi antenna, Discone, Helical, Bow-Tie, and Horn antennas [1-5] have demonstrated wideband performance. However, these designs often come with compromises—large physical footprints, complex feed mechanisms, and in many cases, lower directional gain. Except for the horn antenna, most of these are unsuitable for compact or handheld implementations, making them less ideal for systems where portability and high directionality are critical.

A key example of such systems includes handheld RF guns, which are deployed in applications like radar detection,

jamming, electronic warfare, and short-range point-to-point communication. These systems demand antennas that are lightweight, directional, and capable of handling significant power, all while maintaining a compact footprint. Traditional broadband antennas fall short in these scenarios due to their bulky form factors and structural complexity.

To meet these requirements, folded dipole antennas [1-5] have emerged as a practical and effective alternative. Known for their compact design, directional radiation pattern, and efficient current distribution, these antennas also offer good impedance matching. However, a major limitation persists—restricted bandwidth. Conventional patch-type folded dipole antennas typically offer a bandwidth between 5–10%, limiting their applicability in systems requiring wideband performance. On the other hand, wire-type folded dipoles, while slightly better in bandwidth, are still too large and mechanically complex for integration into compact platforms.

Addressing this trade-off between bandwidth and miniaturization has become a key area of current research. Recent efforts are directed at enhancing the bandwidth of patch-type folded dipole antennas through techniques such as slot loading, parasitic coupling, substrate engineering, and feed structure optimization—all aiming to push performance without compromising compactness.

In this work, a wideband planar folded dipole antenna is presented, designed on a 0.6 mm thick FR4 substrate and optimized using ANSYS HFSS. The antenna operates across the 5–6 GHz band, delivering a 500 MHz bandwidth (7.64%), 5 dB gain, and a beamwidth of $30^\circ \times 30^\circ$. The design effectively balances compactness, performance, and bandwidth, making it a strong candidate for next-generation wireless communication and radar systems.

The remaining sections of this manuscript are organized as follows:

Section II describes the design methodology of the proposed folded dipole antenna.

Section III presents the simulation results and discussion, analysing key antenna parameters such as return loss, gain, bandwidth, and radiation pattern.

Finally, Section IV concludes the research work and highlights the key findings and scope for future improvement.

II. ANTENNA DESIGN METHODOLOGY

The design methodology of the proposed antenna is divided into two sections. The first section of the design methodology highlights the performance of folded dipole patch antenna (FPDA) using FR4 substrate and the second section describes Fishbone antenna. The top and side view of the proposed FPDA design comprises of folded dipole patch on the top of substrate material (FR4) as illustrated in Fig. 1 (a) and 1 (b) respectively. FPDA is fabricated on a $L \times W \times h$. L represents length of FPDA while W represents width and h represents the thickness of the substrate[13]. The designed dimensions of the proposed FPDA is specified in table 1. The proposed FPDA is designed at 5.5 GHz resonance frequency of 5-6 GHz band. The design of the FPDA is structured to achieve high gain, wide bandwidth (5-6 GHz), and narrow beamwidth ($30^\circ \times 30^\circ$) while maintaining a compact and efficient structure. The methodology comprises organized design stages, including shape selection, substrate and conductive material choice, and performance optimization using ANSYS HFSS.

A. Geometry and Structure of FPDA:

The FPDA consists of the tapered feed section provides impedance matching and smooth signal transition as shown in Fig.1 (a). The length of FPDA is optimized using ANSYS HFSS for 5.5 GHz resonance frequency. This is center frequency of 5-6 GHz fast wireless communication band.

The antenna structure is designed using FR4 epoxy substrate, which offers a balance between mechanical durability and electromagnetic performance. The metallic elements are modeled using perfect electric conductors (PECs) to simulate real-world conductive properties.

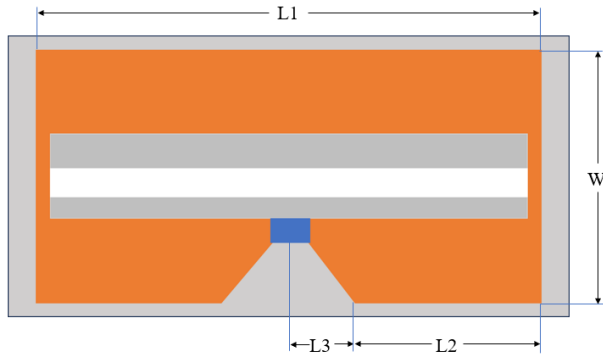


Fig. 1 (a)



Fig. 1 (b)

Figure 1: Folded Patch Type Dipole (FPDA) (a) Top view and (b) Side view

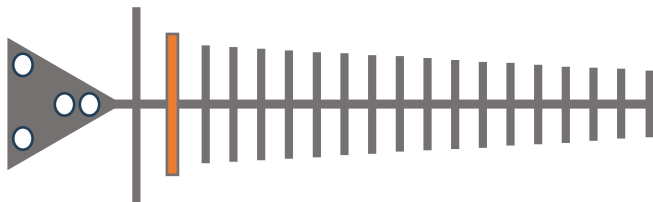


Figure 2: FPDA with parasitic element.

Table 1 Design Specification

Design Parameters	Specification
Resonance Frequency (GHz)	5.5 GHz (5 GHz – 6 GHz)
Bandwidth	500 MHz
Gain	5dB
Power Handling	100W
Beamwidth (HPBW)	$30^\circ \times 30^\circ$
Polarization	Linear
Substrate Material	FR4 epoxy

Table 2 Design Dimensions

Design Parameters	Dimensions
L1	0.58λ
L2	0.21λ
L3	0.05λ
W	0.29λ

The FPDA dimensions has been optimized through parametric analysis in ANSYS HFSS[15], ensuring an optimal balance between length and width, gain, radiation efficiency and radiation pattern of FPDA. The design is simulated in ANSYS HFSS, following these steps:

- Geometry Modeling: Creating a 3D structure based on theoretical calculations.
- Material Assignment: Defining dielectric and conductive properties.
- Boundary Conditions and Excitation: Implementing lumped ports for signal feed.
- Meshing and Solver Setup: Ensuring high-accuracy results through adaptive meshing.
- Performance Analysis: Evaluating return loss (S11), radiation pattern, gain, and directivity.

Similarly the optimization of FPDA are as follows,

- Impedance Matching: Adjusting feedline width and tapering for better S11 performance.
- Gain Enhancement: Optimizing the spacing and length of parasitic elements.
- Beamwidth Control: Refining the fishbone element distribution to maintain a narrow beam.

This methodology ensures that the proposed FPDA meets the required specifications for high-frequency applications, making it suitable for modern radar and wireless communication systems.

B. Geometry and Specification of Fishbone Antenna:

The fishbone antenna is as shown in Fig. 2. This antenna consist of FPDA along with reflector and director as a parasitic element as shown in Fig. 2. The optimization of FPDA is explained in section A. But, optimization of FPDA along with parasitic elements is still in process for better impedance matching as well as excellent radiation characteristics.

III. RESULTS AND DISCUSSION

The proposed FPDA is simulated using ANSYS HFSS[15] to evaluate its performance in terms of return loss (S_{11}), gain, radiation pattern, and half-power beamwidth (HPBW) etc. The proposed design simulated using HFSS software considering the centre frequency of 5.5 GHz from the starting frequency of 4 GHz and end frequency 7 GHz with a step size of 0.1 GHz. The fabrication of proposed FPDA is in process using printed circuit boarded (PCB) technology on 0.6mm FR4 substrate. The results validate the antenna's efficiency and suitability for high-frequency wireless and radar applications.

A. Return Loss (S_{11}) and Impedance Matching:

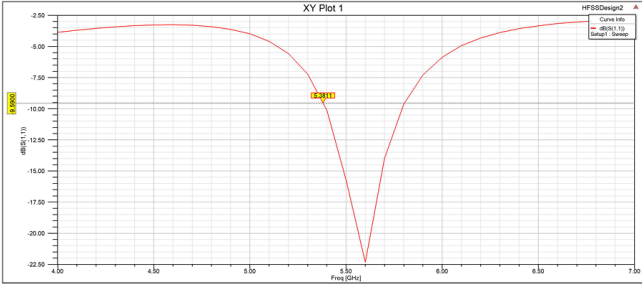


Figure 3: Return Loss (S_{11}) and vs. Frequency.

Fig. 3 shows the frequency vs return loss (S_{11}) response of the proposed FPDA. This graph indicates how much input power is reflected due to impedance mismatch. FPDA has a simulated impedance bandwidth of 420 MHz (5.38 – 5.8 GHz, 7.64 %). The return loss values below –10 dB are considered ideal for efficient power transfer from the transmission line to the FPDA. In the proposed FPDA, the simulated S_{11} remains below –15 dB across the 5–6 GHz band, confirming strong impedance matching throughout the operating range.

A minimum return loss of –25 dB has been notice at 5.5 GHz highlights excellent matching at the center frequency, with negligible reflection and maximum power transfer[6-7]. This result validates the antenna's suitability for wideband wireless and radar applications, where stable impedance characteristics are critical.

B. Frequency vs Impedance Variation:

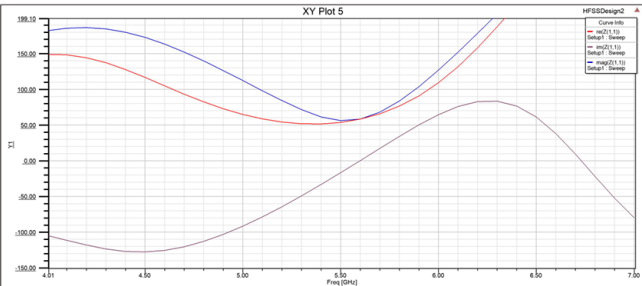


Figure 4: Graph of frequency vs impedance variation.

The Fig. 4 shows the graph of frequency vs impedance variation of the proposed FPDA. This graph indicates that over the frequency of 420 MHz antenna maintains impedance closed to 50 ohm.

C. Gain and Directivity:

Gain reflects the antenna's ability to direct radiated energy in a specific direction. For the proposed folded dipole antenna, the simulated gain varies across the band, reaching a peak of 5 dB at 5.5 GHz. This value confirms stable and efficient radiation within the desired frequency range.

The periodic parasitic structure of the antenna enhances directivity, helping focus the radiation pattern and reduce unwanted side lobes. This makes the design well-suited for directional wireless and radar applications where signal strength and focused beams are essential.

D. VSWR Performance Evaluation:

The voltage standing wave ratio (VSWR) is a critical parameter for evaluating the efficiency of power transmission from the feed line to the antenna. A VSWR value below 2 indicates acceptable impedance matching, where most of the input power is radiated rather than reflected.

The simulated VSWR curve for the proposed folded dipole antenna shows that the VSWR remains below 2 throughout the 5.38 - 5.8 GHz range, which aligns with the antenna's measured bandwidth of 420 MHz. Notably, the VSWR dips closer to 1.2 at the resonant frequency of 5.5 GHz, confirming excellent impedance matching and minimal reflection loss at the design frequency.

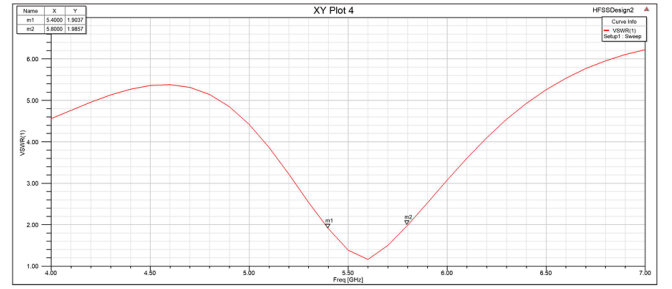


Figure 5: Simulated VSWR vs. Frequency plot of the proposed FPDA.

This result supports the previously observed return loss of –22 dB at 5.5 GHz, reinforcing the antenna's capability to operate efficiently within the intended ultra-wideband range. Such performance makes the antenna highly suitable for practical wireless communication and radar systems, where power transfer efficiency is essential[11].

E. 2D Radiation Pattern and Half-Power Beamwidth:

The 2D radiation pattern of the proposed FPDA is as shown in Fig. 5. The E plane & H plane pattern along with its cross polarization is plotted in Fig. 5. This pattern describes how the antenna radiates energy in space. The HPBW ($30^\circ \times 30^\circ$) ensures high directivity and minimal side lobes.

The antenna exhibits a narrow beamwidth, making it suitable for directional communication and radar applications[9].

The main lobe is well-defined, and side lobes are suppressed, improving radiation efficiency.

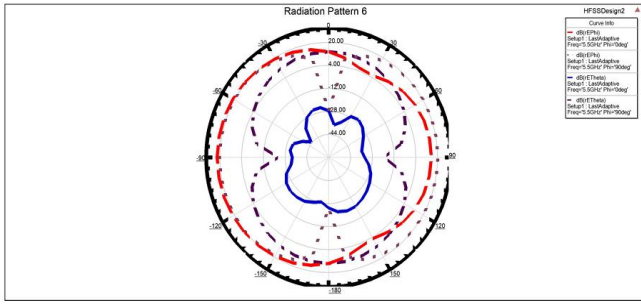


Figure 4: Radiation Pattern at 5.5 GHz

F. Polarization and Efficiency:

The antenna maintains linear polarization, ensuring consistent signal reception and transmission[13].

Radiation efficiency is above 85%, confirming minimal energy losses.

IV. CONCLUSION

This paper presents the design and analysis of a FPDA operating in the 5-6 GHz ultra-wideband frequency range with a gain of 12-5 dB and a narrow beamwidth of $30^\circ \times 30^\circ$. The antenna is optimized for high directivity, efficient power handling (100W), and stable radiation characteristics, making it suitable for modern wireless communication and radar applications.

The proposed design is simulated using ANSYS HFSS[15], and the results confirm excellent impedance matching, low return loss, and high gain performance. The integration of a tapered feed and periodic parasitic elements enhances beamforming capabilities, ensuring efficient energy radiation.

The FPDA's compact structure, linear polarization, and wideband characteristics offer significant advantages over conventional broadband antennas, making it a strong candidate for next-generation high-frequency applications. Future work may focus on fabrication and experimental validation to further refine and validate simulation results.

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